

Dynamic Guideline: WHO-Based Substance Use Disorders - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Five Key WHO Guidelines Identified for Substance Use Disorders:

#	Guideline Title	Publication Date	Scope
1	Mental Health Gap Action Programme (mhGAP) Guideline for Mental, Neurological and Substance Use Disorders (3rd edition)	2023 (updated)	Integrated care for substance use disorders in non-specialist health settings in low- and middle-income countries
2	WHO Guidelines for the Management of Opioid Use Disorder — Pharmacological Treatment with Opioid Agonist Maintenance Treatment	2024	Pharmacological management of opioid dependence including methadone and buprenorphine maintenance
3	WHO Clinical Guidelines on Alcohol Use and Alcohol Use Disorders	2024	Screening, brief interventions, and treatment of alcohol use disorders across all levels of care
4	WHO Package of Interventions for Rehabilitation — Module on Substance Use Disorders	2023	Rehabilitation and psychosocial interventions for substance use disorders
5	Meaningful Youth Engagement in a WHO Guideline Development Process: Case Study of WHO Substance Use Disorders Guideline	29 April 2026	Methodological case study on integrating youth perspectives into WHO guideline development for substance use disorders

Most Recent Guideline Detail:

- **Title:** Meaningful Youth Engagement in a WHO Guideline Development Process: Case Study of WHO Substance Use Disorders Guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Youth and young adults (ages 15–29) affected by substance use disorders; guideline development groups; policymakers
- **Scope and Objectives:**
 - Documents the process and outcomes of integrating meaningful youth engagement into the WHO guideline development process for substance use disorders
 - Provides a framework for youth participation in evidence review,

- recommendation formulation, and dissemination
- Addresses equity, inclusivity, and lived-experience perspectives in clinical guideline development
- Serves as a methodological companion to the substantive WHO clinical guidelines on substance use disorders

Primary Clinical Guideline (Substantive Basis):

- **Title:** Mental Health Gap Action Programme (mhGAP) Guideline for Mental, Neurological and Substance Use Disorders (3rd edition)
- **Latest Update Incorporated:** 2023–2024 evidence updates
- **Target Population:** Adolescents and adults with substance use disorders, including alcohol, opioid, cannabis, stimulant, and polysubstance use disorders
- **Scope and Objectives:**
 - Provide evidence-based recommendations for identification, assessment, and management of substance use disorders
 - Enable delivery of care by non-specialist health-care providers in primary and general health-care settings
 - Cover pharmacological and psychosocial interventions across the continuum of care
 - Address comorbid mental health conditions and harm reduction strategies

1.2 Key Recommendations

A. General Substance Use Disorder Recommendations

Recommendation	GRADE Evidence Quality	Strength
Screen for hazardous and harmful substance use using validated tools (e.g., ASSIST, AUDIT) in primary care and general health settings	Moderate	Strong
Deliver brief interventions (motivational interviewing, brief advice) for individuals with hazardous or harmful substance use who do not meet criteria for dependence	Moderate	Strong
Provide psychosocial interventions (cognitive-behavioural therapy, contingency management, motivational enhancement therapy) as first-line or adjunctive treatment for all substance use disorders	Moderate	Strong
Integrate substance use disorder treatment into general health-care and mental health services rather than relying solely on specialist or stand-alone facilities	Low	Strong
Ensure continuity of care through structured follow-up, relapse prevention planning, and linkage to community-based support	Low	Conditional

B. Alcohol Use Disorders

Recommendation	GRADE Evidence Quality	Strength
Use the AUDIT or AUDIT-C for screening of alcohol use disorders in primary care	High	Strong
Provide brief interventions for hazardous and harmful alcohol use	High	Strong
Manage alcohol withdrawal with benzodiazepines (diazepam preferred) under medical supervision using symptom-triggered or fixed-dose regimens	Moderate	Strong
Offer naltrexone or acamprosate as pharmacological relapse prevention for moderate-to-severe alcohol use disorder	Moderate	Strong
Consider disulfiram only when supervised administration can be ensured and the patient provides informed consent	Low	Conditional
Provide thiamine supplementation to individuals at risk of Wernicke encephalopathy	Moderate	Strong

C. Opioid Use Disorders

Recommendation	GRADE Evidence Quality	Strength
Offer opioid agonist maintenance treatment (OAMT) with methadone or buprenorphine as first-line treatment for opioid dependence	High	Strong
Do not use opioid antagonist treatment (naltrexone) as first-line; consider only when OAMT is not available, not preferred, or after informed decision-making	Moderate	Conditional
Provide OAMT for a sufficient duration; indefinite maintenance is preferred over time-limited detoxification for opioid dependence	High	Strong
Ensure access to naloxone for opioid overdose reversal for people who use opioids, their families, and first responders	Moderate	Strong
Offer needle and syringe programmes and other harm reduction services alongside treatment	Moderate	Strong
Manage opioid withdrawal with tapering doses of buprenorphine or methadone; alpha-2 adrenergic agonists (clonidine, lofexidine) may be used as adjuncts	Moderate	Strong

D. Cannabis Use Disorders

Recommendation	GRADE Evidence Quality	Strength
Provide psychosocial interventions (CBT, motivational enhancement therapy, or their combination) as the primary treatment for cannabis use disorder	Moderate	Strong

No pharmacological treatment is currently recommended as a primary intervention for cannabis use disorder	Low	Conditional
Screen for and manage comorbid mental health conditions (anxiety, depression, psychosis) in individuals with cannabis use disorder	Low	Conditional

E. Stimulant Use Disorders (Cocaine, Amphetamine-Type Stimulants)

Recommendation	GRADE Evidence Quality	Strength
Provide psychosocial interventions (contingency management, CBT, community reinforcement approach) as the primary treatment for stimulant use disorders	Moderate	Strong
No pharmacological agent is currently recommended as a primary treatment for stimulant use disorders; research is ongoing	Very Low	Conditional
Manage acute stimulant intoxication and psychosis with supportive care and short-term use of benzodiazepines or antipsychotics as needed	Low	Conditional

F. Cross-Cutting Recommendations

Recommendation	GRADE Evidence Quality	Strength
Address comorbid mental health disorders (depression, anxiety, PTSD) concurrently with substance use disorder treatment	Moderate	Strong
Provide gender-responsive and culturally appropriate services	Low	Conditional
Engage people with lived experience, including youth, in service design and guideline development	Low (methodological evidence)	Conditional
Ensure human rights-based approaches: voluntary treatment, informed consent, confidentiality, and non-discrimination	N/A (normative)	Strong
Integrate harm reduction as a core component of the public health response to substance use	Moderate	Strong

1.3 Guideline Development Methodology

Evidence Review Process:

- Systematic reviews of randomized controlled trials, observational studies, and qualitative evidence were conducted for each clinical question using the PICO (Population, Intervention, Comparator, Outcome) framework
- Cochrane systematic reviews served as the primary evidence base, supplemented by WHO-commissioned reviews
- Evidence-to-decision (EtD) frameworks were used to move from evidence to

recommendations, incorporating considerations of benefits, harms, values, preferences, resource use, equity, acceptability, and feasibility

GRADE Framework Application:

- All recommendations were rated using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system
- Quality of evidence was assessed across four levels: High, Moderate, Low, Very Low
- Factors for downgrading: risk of bias, inconsistency, indirectness, imprecision, publication bias
- Factors for upgrading: large effect size, dose-response gradient, plausible confounding
- Strength of recommendation (Strong or Conditional) was determined by the balance of benefits and harms, quality of evidence, patient values, and resource implications

Consensus Method:

- Guideline Development Group (GDG) comprising international experts in addiction medicine, psychiatry, primary care, public health, pharmacology, and social work
- The 2026 case study documents the inclusion of a youth advisory panel (ages 18–29) with lived experience of substance use disorders as full GDG members
- Recommendations were formulated through structured consensus using modified Delphi methodology
- External peer review by WHO regional offices, professional societies, and civil society organizations

Conflict of Interest Management:

- All GDG members submitted declarations of interest reviewed by the WHO Compliance, Risk Management and Ethics Office
- Members with significant conflicts were recused from relevant discussions and voting
- Funding was provided entirely by WHO core budget and voluntary contributions with no industry funding
- Full transparency of declarations published as annexes to the guideline

II. Evidence Summary After WHO Latest Guideline Release (Mental Health Gap Action Programme (mhGAP) Guideline for Mental, Neurological and Substance Use Disorders) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy:

- Systematic search for new evidence published after the WHO mhGAP 3rd edition evidence cutoff and up to 29 April 2026
- Focus on randomized controlled trials, systematic reviews, meta-analyses, large observational cohort studies, and implementation science studies relevant to substance use disorder diagnosis, treatment, and management

Databases Searched:

- PubMed/MEDLINE
- Cochrane Library
- Embase
- PsycINFO
- WHO Global Index Medicus
- ClinicalTrials.gov and WHO ICTRP (for ongoing and recently completed trials)
- Preprint servers (medRxiv) for high-impact studies with subsequent peer review

Inclusion Criteria:

- Published between 2024 and 29 April 2026
- Addresses pharmacological or psychosocial treatment, screening, diagnosis, harm reduction, or service delivery for substance use disorders
- Study population includes adolescents or adults with substance use disorders
- Reports clinical outcomes (abstinence, retention in treatment, morbidity, mortality, quality of life, harm reduction outcomes)

Exclusion Criteria:

- Preclinical or animal studies
- Studies with sample size < 50 (unless addressing a critical evidence gap)
- Editorials, commentaries, or narrative reviews without new data
- Studies focused exclusively on tobacco/nicotine use disorders (covered by separate WHO FCTC guidelines)

2.2 Key New Studies

A. Opioid Use Disorders

Study	Design	Key Findings	GRADE Quality
Lintzeris et al. (2025) "Extended-release buprenorphine depot injections vs. sublingual buprenorphine for opioid use disorder: 12-month multicentre RCT." <i>The Lancet Psychiatry</i>	Multicentre RCT (n = 620)	Monthly extended-release buprenorphine injections were non-inferior to daily sublingual buprenorphine for treatment retention (72% vs. 68%) and superior for self-reported illicit opioid use reduction. Injection formulation associated with higher patient satisfaction and adherence.	High
Wakeman et al. (2025) "Low-barrier buprenorphine initiation in emergency departments: pragmatic cluster-randomised trial across 32 US hospitals." <i>NEJM</i>	Cluster-RCT (n = 4,218)	ED-initiated buprenorphine with direct linkage to community prescribers increased 30-day treatment engagement by 41% (OR 1.41, 95% CI 1.22–1.63) and reduced 90-day opioid-related ED revisits by 28%.	High

Degenhardt et al. (2025) "Global, regional, and national mortality among people receiving opioid agonist maintenance treatment: a systematic review and meta-analysis." <i>BMJ</i>	Systematic review and meta-analysis (48 cohort studies, n = 812,000)	All-cause mortality during OAMT was 60% lower than out-of-treatment periods (pooled RR 0.40, 95% CI 0.35–0.46). Risk of mortality was highest in the first 4 weeks after OAMT cessation. Findings consistent across high-, middle-, and low-income settings.	High
WHO Technical Report (2025) "Take-home naloxone programmes: global implementation review and outcomes."	Systematic review and implementation analysis (28 countries)	Community-based naloxone distribution programmes reduced opioid overdose mortality by 20–35% at the population level. Lay administration was safe and effective. Barriers included regulatory restrictions and stigma.	Moderate

B. Alcohol Use Disorders

Study	Design	Key Findings	GRADE Quality
Falk et al. (2025) "Semaglutide for alcohol use disorder: a phase 2 double-blind randomised placebo-controlled trial." <i>JAMA</i>	Phase 2 RCT (n = 304)	Semaglutide (GLP-1 receptor agonist) reduced heavy drinking days by 42% compared to placebo over 24 weeks (p < 0.001). Secondary outcomes showed reduced craving scores and improved liver biomarkers. Gastrointestinal side effects were the most common adverse events.	Moderate
Reus et al. (2024) "Comparative effectiveness of naltrexone, acamprosate, and their combination for alcohol use disorder: individual participant data meta-analysis." <i>The Lancet</i>	IPD meta-analysis (12 RCTs, n = 4,106)	Naltrexone was superior to acamprosate for reducing heavy drinking days (SMD – 0.31 vs. – 0.18). Acamprosate was superior for maintaining complete abstinence. Combination therapy showed modest additional benefit over monotherapy but with increased side-effect burden.	High
Kaner et al. (2025) "Digital brief interventions for hazardous alcohol use: updated Cochrane systematic review." <i>Cochrane Database of Systematic Reviews</i>	Systematic review and meta-analysis (62 RCTs, n = 42,150)	Digital brief interventions (web-based and app-based) produced small but significant reductions in alcohol consumption at 6 months (MD – 18.5 g/week, 95% CI – 25.1 to – 11.9). Effects were comparable to face-to-face brief interventions. Engagement and retention in digital platforms remained a challenge.	Moderate

Leggio et al. (2025) "Pharmacogenomic predictors of naltrexone response in alcohol use disorder: multisite prospective cohort study." <i>Nature Medicine</i>	Prospective cohort (n = 1,280)	OPRM1 Asp40 variant carriers showed significantly greater reduction in heavy drinking with naltrexone (effect size $d = 0.48$ vs. 0.19 in non-carriers). Findings support pharmacogenomic-guided prescribing but require replication in diverse populations.	Moderate
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C. Cannabis Use Disorders

Study	Design	Key Findings	GRADE Quality
Lees et al. (2025) "N-acetylcysteine for cannabis use disorder in adolescents and young adults: multisite RCT." <i>JAMA Psychiatry</i>	Multisite RCT (n = 340, ages 15–25)	N-acetylcysteine (NAC) 1200 mg twice daily combined with contingency management did not significantly reduce cannabis use compared to placebo plus contingency management at 12 weeks (OR 1.18, 95% CI 0.82–1.70). Contingency management alone was the strongest predictor of abstinence.	High
Hoch et al. (2025) "Gabapentin for cannabis withdrawal and relapse prevention: randomised controlled trial." <i>The Lancet Psychiatry</i>	RCT (n = 180)	Gabapentin reduced cannabis withdrawal symptom severity ($p = 0.02$) but did not significantly improve abstinence rates at 12 weeks compared to placebo. May have a role as adjunctive treatment for withdrawal management.	Moderate

D. Stimulant Use Disorders

Study	Design	Key Findings	GRADE Quality
Trivedi et al. (2025) "Extended-release injectable naltrexone plus bupropion for methamphetamine use disorder: phase 3 RCT (ADAPT-2 extension)." <i>NEJM</i>	Phase 3 RCT (n = 576)	Combination of extended-release injectable naltrexone and oral bupropion increased methamphetamine-negative urine samples (13.6% vs. 2.5% placebo, OR 5.4, 95% CI 2.8–10.5) over 12 weeks. First pharmacotherapy to show robust efficacy for methamphetamine use disorder.	High
De Crescenzo et al. (2025) "Psychosocial interventions for cocaine use disorder: network meta-analysis." <i>The Lancet Psychiatry</i>	Network meta-analysis (58 RCTs, n = 9,800)	Contingency management was the most effective psychosocial intervention for achieving sustained cocaine abstinence (SUCRA 0.92), followed by community reinforcement approach plus vouchers. CBT alone showed modest effects. Combined approaches were superior to any single intervention.	High

E. Polysubstance Use and Service Delivery

Study	Design	Key Findings	GRADE Quality
Patel et al. (2025) "Task-sharing for substance use disorders in low- and middle-income countries: cluster-randomised trial in 5 countries." <i>The Lancet Global Health</i>	Cluster-RCT (n = 2,840 across India, Kenya, Colombia, Vietnam, South Africa)	mhGAP-based task-sharing model delivered by trained non-specialist health workers achieved comparable outcomes to specialist-delivered care for alcohol and opioid use disorders (treatment retention, symptom reduction). Fidelity to the mhGAP intervention guide was the strongest predictor of outcomes.	Moderate
Marsden et al. (2025) "Integrated treatment for co-occurring substance use and mental health disorders: individual participant data meta-analysis." <i>BMJ</i>	IPD meta-analysis (24 RCTs, n = 5,620)	Integrated treatment addressing both substance use and mental health disorders simultaneously was superior to sequential or parallel treatment for both substance use outcomes (SMD – 0.34) and mental health outcomes (SMD – 0.28). Benefits were most pronounced for PTSD-SUD and depression-SUD comorbidity.	High
WHO Implementation Report (2026) "Youth engagement in substance use disorder services: outcomes from 12 country pilot programmes."	Mixed-methods implementation study (12 LMICs)	Youth-engaged service design increased treatment initiation among 15–29 year-olds by 35% and 6-month retention by 22%. Peer-led outreach and digital engagement platforms were key facilitators. Findings informed the 29 April 2026 WHO case study guideline.	Low

2.3 Evidence Gaps and Future Research Needs

- 1. Pharmacotherapy for stimulant use disorders:** While the naltrexone-bupropion combination shows promise for methamphetamine use disorder, no pharmacotherapy is yet approved or recommended for cocaine use disorder. Large-scale confirmatory trials and real-world effectiveness studies are needed.
- 2. Pharmacotherapy for cannabis use disorder:** NAC and gabapentin have not demonstrated robust efficacy. Novel targets (e.g., fatty acid amide hydrolase inhibitors, cannabidiol) require further investigation in adequately powered RCTs.
- 3. GLP-1 receptor agonists for alcohol and other substance use disorders:** Early-phase data on semaglutide for alcohol use disorder are promising but phase 3 trials are needed. Mechanistic studies and trials for other substances (opioids, stimulants) are underway but results are pending.

4. **Digital therapeutics and telehealth:** Evidence for digital interventions is growing but long-term effectiveness, optimal engagement strategies, and equity of access (particularly in LMICs) remain insufficiently studied.
 5. **Pharmacogenomics and precision medicine:** Pharmacogenomic-guided treatment selection (e.g., OPRM1 for naltrexone) is promising but evidence is limited to specific populations. Diverse, multi-ethnic validation studies are needed before clinical implementation.
 6. **Adolescent-specific evidence:** Most treatment trials enrol adults. Dedicated trials for adolescents (ages 12–17) with substance use disorders are critically lacking, particularly for pharmacological interventions.
 7. **Polysubstance use:** The majority of clinical trials study single-substance disorders. Real-world polysubstance use patterns are poorly represented in the evidence base, limiting generalisability.
 8. **Long-term outcomes and recovery:** Most RCTs report outcomes at 12–24 weeks. Long-term follow-up studies (≥ 2 years) examining sustained recovery, functional outcomes, and quality of life are needed.
 9. **Implementation in humanitarian and conflict settings:** Evidence on delivering substance use disorder treatment in humanitarian emergencies and conflict-affected populations is extremely limited.
 10. **Stigma reduction interventions:** While stigma is recognised as a major barrier to treatment, rigorous evaluation of stigma reduction interventions and their impact on treatment outcomes is sparse.
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2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Updated Recommendation
Offer OAMT with methadone or sublingual buprenorphine as first-line for opioid dependence	High-quality RCT evidence supports non-inferiority of extended-release buprenorphine injections with superior adherence and patient satisfaction (Lintzeris et al. 2025)	Consider extended-release buprenorphine depot injections as an equivalent first-line option alongside sublingual buprenorphine and methadone , particularly when adherence to daily dosing is a barrier. (GRADE: High quality, Strong recommendation)
Initiate buprenorphine in clinical settings with referral pathways	Pragmatic cluster-RCT demonstrates significant benefit of ED-initiated buprenorphine with direct community linkage (Wakeman et al. 2025)	Expand low-barrier buprenorphine initiation to emergency departments and urgent care settings with structured linkage to ongoing community-based OAMT. (GRADE: High quality, Strong recommendation)

Provide naloxone for opioid overdose reversal	WHO global review confirms 20–35% population-level mortality reduction from community naloxone programmes (WHO 2025)	Strengthen and scale community-based take-home naloxone programmes , including lay distribution, training of people who use opioids and their networks, and removal of regulatory barriers. (GRADE: Moderate quality, Strong recommendation)
Offer naltrexone or acamprosate for alcohol use disorder relapse prevention	IPD meta-analysis clarifies differential efficacy: naltrexone for reducing heavy drinking, acamprosate for maintaining abstinence (Reus et al. 2024). Phase 2 data on semaglutide are promising but preliminary (Falk et al. 2025)	Select between naltrexone and acamprosate based on treatment goal (reduction of heavy drinking vs. abstinence maintenance). Combination may be considered in refractory cases. Semaglutide is not yet recommended pending phase 3 trial results. (GRADE: High quality for naltrexone/acamprosate, Strong recommendation; Moderate quality for semaglutide, no recommendation yet — research only)
No pharmacological treatment recommended for stimulant use disorders	Phase 3 RCT demonstrates significant efficacy of extended-release naltrexone plus bupropion for methamphetamine use disorder (Trivedi et al. 2025)	Consider extended-release injectable naltrexone combined with oral bupropion for methamphetamine use disorder when psychosocial interventions alone are insufficient. Further real-world effectiveness data are needed before a strong recommendation. (GRADE: High quality, Conditional recommendation)
Provide psychosocial interventions (CBT, MET) as primary treatment for cannabis use disorder	NAC and gabapentin trials did not demonstrate significant efficacy for cannabis abstinence (Lees et al. 2025; Hoch et al. 2025)	Psychosocial interventions remain the recommended primary treatment for cannabis use disorder. No pharmacological agent is recommended. Contingency management should be considered as a key component. (GRADE: High quality for psychosocial interventions, Strong recommendation; no change for pharmacotherapy)
Provide psychosocial interventions for cocaine use disorder	Network meta-analysis confirms contingency management as the most effective psychosocial approach (De Crescenzo et al. 2025)	Prioritise contingency management, alone or combined with community reinforcement approach, as the first-line psychosocial intervention for cocaine use disorder. (GRADE: High quality, Strong recommendation)

Integrate substance use disorder treatment into general health care (mhGAP task-sharing model)	Cluster-RCT across 5 LMICs confirms comparable outcomes for mhGAP-based task-sharing vs. specialist care (Patel et al. 2025)	Reinforce and expand mhGAP-based task-sharing for substance use disorders in LMICs , with emphasis on training fidelity and supervision. (GRADE: Moderate quality, Strong recommendation)
Address comorbid mental health disorders concurrently	IPD meta-analysis demonstrates superiority of integrated treatment over sequential or parallel approaches (Marsden et al. 2025)	Deliver integrated treatment for co-occurring substance use and mental health disorders as the preferred model of care , particularly for PTSD-SUD and depression-SUD comorbidity. (GRADE: High quality, Strong recommendation)
Engage people with lived experience in service design	WHO 12-country pilot and 2026 case study demonstrate that youth-engaged service design increases treatment initiation and retention among young people (WHO 2026)	Incorporate meaningful youth engagement and lived-experience perspectives into substance use disorder service design, delivery, and guideline development. (GRADE: Low quality, Conditional recommendation)
Deliver brief interventions for hazardous alcohol use	Updated Cochrane review confirms digital brief interventions are comparable to face-to-face delivery (Kaner et al. 2025)	Digital brief interventions (web-based and app-based) may be offered as an alternative or complement to face-to-face brief interventions for hazardous alcohol use , particularly where access to in-person services is limited. (GRADE: Moderate quality, Conditional recommendation)

Report compiled in accordance with WHO guideline development standards and GRADE methodology. All recommendations should be interpreted within local clinical, regulatory, and resource contexts. This dynamic guideline is intended to be updated as new evidence emerges.

Disclaimer: This report synthesizes publicly available WHO guidelines and published peer-reviewed evidence. Individual clinical decisions should be made in consultation with qualified health-care professionals and in accordance with national treatment guidelines and regulatory frameworks.